NEET Chemistry — Thermodynamics

Ultra Premium Practice Sheet (20 Questions)

Q1. For an isothermal process, ΔU is:

A. Zero

B. Positive

C. Negative

D. Depends

Q2. Work done in isobaric process is:

A. PdV

B. $P\Delta V$

C. $V\Delta P$

D. None

Q3. First law: ΔU =

A. q + w

B. q - w

C. w - q

D. q × w

Q4. Enthalpy change at constant pressure equals:

A. Heat

B. Work

C. Energy

D. Zero

Q5. Which is state function?

A. Work

B. Heat

C. Enthalpy

D. Path

Q6. Adiabatic process has:

A. q=0

B. w=0

C. $\Delta U=0$

D. None

Q7. Efficiency of heat engine depends on:

A. Temp
B. Pressure
C. Volume
D. Mass

Q8. $\Delta H = \Delta U + ?$

A. PV
B. $P\Delta V$
C. $V\Delta P$
D. None

Q9. Spontaneous process has ΔG:

A. Negative
B. Positive
C. Zero
D. Infinite

Q10. Entropy measures:

A. Disorder
B. Energy
C. Heat
D. Work

Q11. For ideal gas ΔU depends on:

A. Temp
B. Pressure
C. Volume
D. None

Q12. Endothermic process:

A. Absorbs heat
B. Releases heat
C. No heat
D. None

Q13. Exothermic process:

A. Absorbs heat

B. Releases heat

C. No heat

D. None

Q14. Carnot engine efficiency:

A. 1 - T2/T1

B. T2/T1

C. T1/T2

D. None

Q15. $\Delta G = \Delta H$ - ?

A. $T\Delta S$

B. ΔST

C. TS

D. None

Q16. At equilibrium ΔG:

A. 0

B. Positive

C. Negative

D. Infinite

Q17. Entropy increases in:

A. Expansion

B. Compression

C. Cooling

D. None

Q18. Work done in cyclic process:

A. Zero

B. Max

C. Min

D. Depends

Q19. Heat capacity at constant pressure:

A. C_p

B. C_v

C. R

D. None

Q20. Relation Cp - Cv =

A. R

B. 0

C. 1

D. None

Answer Key

Q1: A
Q2: B
Q3: A
Q4: A
Q5: C
Q6: A
Q7: A
Q8: B
Q9: A
Q10: A
Q11: A
Q12: A
Q13: B
Q14: A
Q15: A
Q16: A
Q17: A
Q18: A
Q19: A
Q20: A